

TOOLS AND RESOURCES

FAST – filamentous actin segmentation tool for quantifying cytoskeletal organization

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ABSTRACT

Studying how actin filaments are assembled into different subcellular structures can provide insights into both physiological processes and the mechanisms of disease. However, quantifying the size, abundance and organization of different classes of actin structure from optical microscopy data remains a challenge. To address this, we developed a deep learning-based tool called the ‘filamentous actin segmentation tool’ (FAST) to accurately and efficiently segment and quantify different classes of actin structure from phalloidin stained confocal microscopy images. We evaluated the performance of this tool to segment and quantify the abundance of different classes of actin structure in different cell lines and with dynamic changes in actin organization using LifeAct–GFP during drug treatments. FAST enables quantification of different classes of actin structure from actin images alone, without the need for specific antibodies against proteins in different actin structures and hence can be a useful tool for researchers studying actin-related pathways involved in cell motility, cancer metastasis and drug development.

KEY WORDS: Actin, Cytoskeleton, Fluorescence microscopy, Segmentation, Deep learning

INTRODUCTION

The actin cytoskeleton plays a crucial role in a range of cellular functions including driving cell motility, determining cell shape and mechanical properties, and separating daughter cells during cytokinesis, to name a few (Dominguez and Holmes, 2011; Fletcher and Mullins, 2010; Pollard, 2016). This broad range of functions is enabled through the assembly of filamentous actin (F-actin) into a variety of specialized structures, each with unique characteristics (Fletcher and Mullins, 2010; Michelot and Drubin, 2011). Common classes of actin structure include lamellipodial and lamellar networks, filopodia, stress fibers and focal adhesions, which are defined by the morphology of the structure, the organization of actin filaments within the structure, and the presence of specific actin regulatory proteins (Desroches and Harris, 2024; Harris et al., 2018). Lamellipodial networks are broad, thin sheet-like extensions that are 0.1–0.2 µm thick and several microns wide. They play a crucial role at the leading edge of

migrating cells and drive cell motility, wound healing and immune cell functions (Arce et al., 2023; Innocenti, 2018). Lamellipodia consist of branched actin filaments formed through activity of the Arp2/3 complex (Suraneni et al., 2012). Filopodia are finger-like protrusions that are 0.1–0.3 µm in diameter and range in length from a few microns to more than 35 µm long (Mattila and Lappalainen, 2008; Schäfer et al., 2011). They act as sensory structures and play an important role in exploratory cell migration (Davenport et al., 1993; Gupton and Gertler, 2007; Mattila and Lappalainen, 2008). Within filopodia, parallel bundles of actin filaments are crosslinked by the members of the fascin protein family with proteins such as myosin X located at filopodial tips (Gupton and Gertler, 2007; Yamashiro et al., 1998). Stress fibers are cytoplasmic bundles of actin filaments of ~0.5–1 µm in diameter and range in length from several microns to tens of microns long (Burridge and Wittchen, 2013). One major function of stress fibers is to generate cellular contractile forces (Lu et al., 2008). Stress fibers contain anti-parallel actin filaments crosslinked by proteins such as members of the α -actinin family, and the contractile activity of stress fibers is driven by myosin II (Tojkander et al., 2012). Some stress fibers terminate at focal adhesions, which are small, ~1–2 µm in diameter, structures located at sites of contact with the surrounding environment (Geiger et al., 2009; Kanchanawong et al., 2010). Focal adhesions contain a number of different proteins that including adapter proteins, such as paxillin and integrins (Askari et al., 2010; Rivelino et al., 2001; Schaller, 2001).

Given that different actin structures fulfill specific functions, quantifying their presence, abundance and localization can yield valuable information about the physiological state of a cell (Desroches and Harris, 2024). Fluorescence microscopy is the tool of choice for characterizing the organization of the actin cytoskeleton both in live cells, by expressing and imaging fluorescent fusion proteins to probes such as LifeAct (Riedl et al., 2008), f-tractin (Lopata et al., 2018), and actin binding domains (Burkel et al., 2007; Harris et al., 2020, 2019), and in fixed cells using fluorescently conjugated phalloidin (Adams and Pringle, 1991; Melak et al., 2017). Furthermore, actin staining with phalloidin can be combined with immunostaining for actin regulatory proteins to assist with identifying a particular class of actin structure. For instance, myosin X is involved in the initiation and extension of filopodia and localizes to filopodial tips. Immunostaining for Myosin X has therefore become a common approach for identifying filopodia in microscopy images. Critically, the efficacy of this approach is highly dependent on the signal to noise ratio and quality of the image that can be obtained with a particular antibody to obtain reliable identification of different classes of actin structure. Another limitation arises from the availability and cross reactivity of antibodies for multiple different classes of actin structures that can be used in tandem, for multi-class identification (i.e. host species, limited number of fluorescence imaging channels).

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Beyond simply identifying actin structures with antibodies against actin regulatory proteins, image analysis methods to segment and quantify the abundance and localization of different classes of actin structure are continually being developed (Desroches and Harris, 2024). For filopodial quantification, tools like *filoVision* (Eddington et al., 2024), *FiloQuant* (Jacquemet et al., 2019) and *Filopodyan* (Urbančič et al., 2017) are used with phalloidin staining to identify F-actin and in some cases with filopodial markers like myosin X. These tools identify the cell boundary using either a deep learning model (*filoVision*) or contour detection in *ImageJ*, which is subsequently used to detect filopodial protrusions based on morphology (*FiloQuant*, *Filopodyan*) or with a filopodial marker (*filoVision*). Another *ImageJ* algorithm, called *FilamentSensor* (Hauke et al., 2023), is suited for detection of stress fiber-related features. Focal adhesions have been detected with imaging tools like *CellProfiler* (Carpenter et al., 2006) and *SFAlab* (Mostert et al., 2023) with thresholding, edge detection and deep learning (Mohamed et al., 2024). Despite these advances, a key challenge is that these tools are typically developed for specific cell types and single classes of actin structure, which constrain their widespread use. To address these challenges, we developed a unique approach using antibody assisted labeling to create high quality ground truth annotations for detecting multiple classes of actin structures. This dataset was used to train a deep learning pipeline, which we refer to as the ‘filamentous actin segmentation tool’ (FAST), to quantify the abundance and properties of actin structure classes.

RESULTS

Antibody-assisted annotation

We sought to develop an image analysis approach that could segment multiple classes of actin structure from a phalloidin (F-actin) image alone, eliminating the need for using multiple antibodies to identify and segment different classes of actin structure at the inference step. Deep learning is well suited to this task and has seen increasing use in optical microscopy image classification and segmentation tasks for images of cells and subcellular structures (Metlek, 2024; Stringer et al., 2021; Melanthota et al., 2022). Training a deep learning algorithm to perform image segmentation requires a high-quality labeled dataset to ensure accurate results. To generate a labeled dataset that can be used to segment different classes of actin structure, we used antibodies targeting actin regulatory proteins to fluorescently label and identify different classes of actin structure. The localization of these antibodies could then be used to guide the annotation of four distinct classes of actin structure in phalloidin images, namely, lamellipodia and lamellar regions, filopodia, stress fibers, and focal adhesions.

We screened through a selection of antibodies that could be used for labeling based on three criteria (Fig. S1). First, we excluded antibodies that failed to produce a significant signal-to-background ratio (actin structure to cytoplasmic or non-specific staining). This characterization was based on colocalization of the signal from the antibody with phalloidin, and the expected structure of interest (Fig. S1). Second, given that we wanted to label several classes of actin structure in the same image for multi-class segmentation, we selected antibodies based on their compatibility with one another (i.e. availability or host species). Third, to minimize fluorescence crosstalk between imaging channels, fluorescently labeled secondary antibodies were chosen to ensure minimal overlap between fluorophore excitation and emission spectra, enabling clear distinction between different classes of actin structure. A summary of the antibodies that

were tested but excluded is provided in Table S1. Based on these selection criteria, F-actin was labeled with phalloidin-iFluor 647 reagent (Abcam, ab176759), paxillin staining was used to label focal adhesions [Abcam, ab32084 with Alexa Fluor® 568 goat anti-rabbit-IgG secondary antibody (Abcam, ab175471)], myosin X staining was used for annotating filopodia [Novus Biologicals, CL8994 with Alexa Fluor® 405 goat anti-mouse-IgG secondary antibody (Life Technologies Inc, A31553)], and lamellipodia were annotated at regions determined as lacking human myosin IIA [GeneTex, GTX33939 staining with goat anti-human-IgG (H+L) Alexa Fluor® 488 (Life Technologies Inc, A11013)]. To generate training datasets, we fixed and stained HeLa cells (see Materials and Methods), which are widely used as an adherent cell model. Following immunostaining, cells were imaged using a 60× Nikon water-immersion objective with a spinning disk confocal microscope (Fig. 1A,B, see Materials and Methods). We collected 297 multichannel images across four biological replicates (~75 images per replicate).

During data labeling, we ensured images contained only single cells by cropping the centermost cell where applicable. To assist with annotation, we performed background subtraction, contrast enhancement, and applied a binary threshold (Fig. S2). We then used the following distinct visual signatures from image channels associated with each class of actin structure as a guide for semi-automated creation of annotation masks (Fig. 1A,B). To obtain lamellipodial and lamellar regions, we leveraged the channel corresponding to myosin II and phalloidin and highlighted the regions at the cell periphery where phalloidin was present, but myosin II was absent. Although this pattern is also applicable for filopodia, we confirmed filopodial presence by using the merged image of myosin X and phalloidin, as filopodial tips typically show strong myosin X localization (Bohil et al., 2006). To detect stress fibers, we used images from the phalloidin channel alone. Note that even though there are multiple types of stress fibers (of dorsal, ventral and transverse arcs) that are seen in our imaging, they were collectively treated as a single class of stress fibers for simplicity. For focal adhesions, we used images from the paxillin channel and highlighted the bright puncta corresponding to focal adhesions on the cell. Masks were generated using a combination of automated and manual annotations, supervised by human-in-the-loop verification (Fig. 1C,D). Manual annotation, verification and correction of the masks were carried out using web-based annotation platform Supervisely (<https://supervisely.com/>, version 6.8.67, accessed in 2023).

In total, 1188 images across the four channels were processed and annotated to create the training dataset.

FAST training and evaluation

To train the model, the phalloidin channel for a single cell was used as the input image, with a mask containing four labels for different classes of actin structure. We used a supervised learning approach and selected an extension of UNet model with enhanced skip connections (UNet++) due to its fast inference speed and demonstrated success with biomedical datasets (Azad et al., 2024; Zhou et al., 2018). This model, equipped with a ResNet-34 encoder (He et al., 2016) (Fig. S3A), was pre-trained on the ImageNet dataset (Deng et al., 2009). The model is configured to accept single-channel input images from phalloidin staining and produce an output segmentation mask with five classes, including the image background and four classes of actin structures of interest. To improve model generalization and reduce overfitting, we randomly transformed our dataset with added noise, cropped images and altered brightness (Fig. S3B). The model was trained using Dice

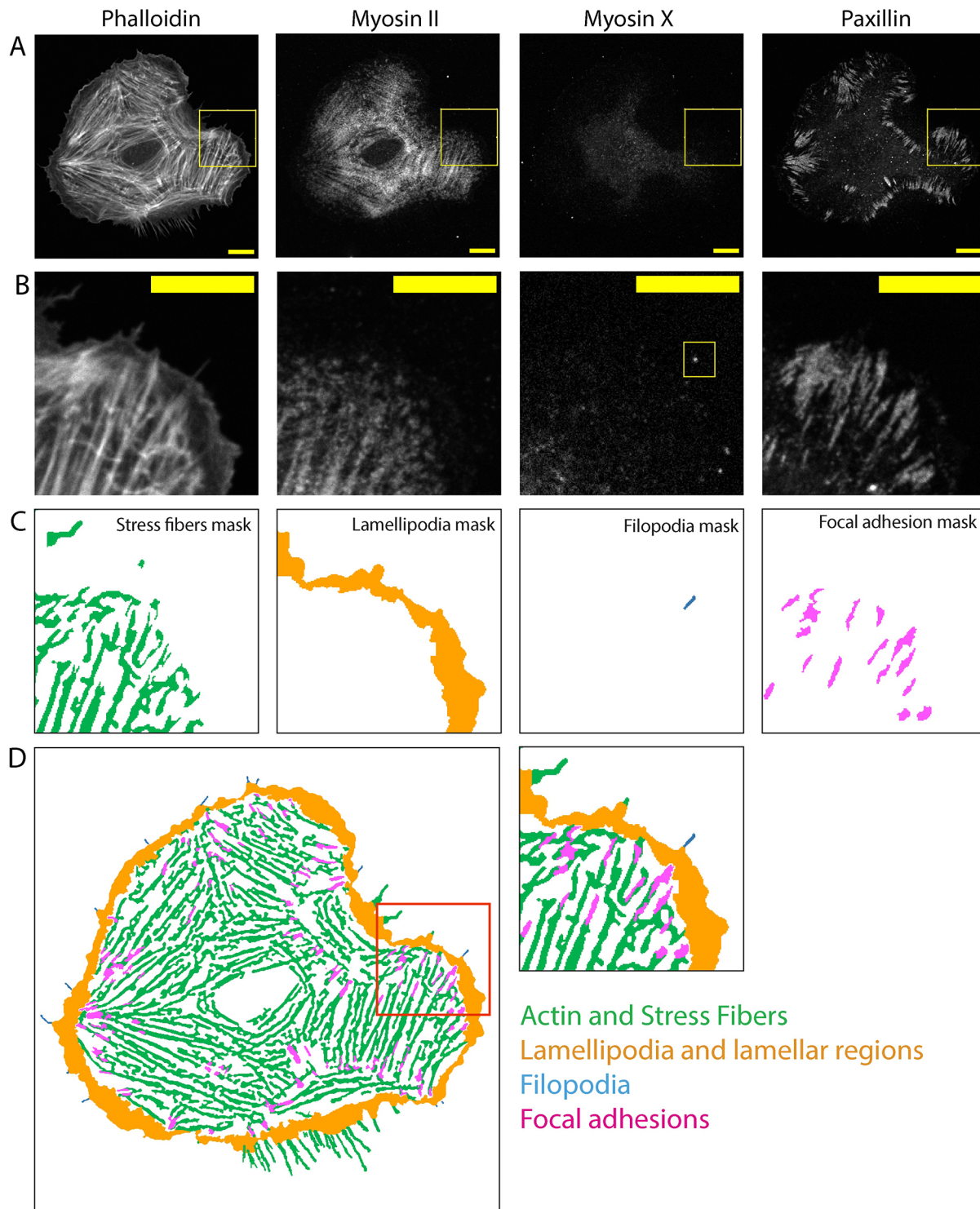


Fig. 1. Antibody assisted labeling. (A) Representative image (of 267 cells used for training from three experimental repeats) of a HeLa cell labeled with phalloidin, myosin II, myosin X and paxillin. (B) Magnified images of highlighted region (yellow box in A). (C) Phalloidin was used to detect actin and stress fibers, the absence of myosin II was used to identify lamellipodia and lamellar regions, the presence of myosin X puncta (yellow box in B) at the end of thin protrusions was used for detecting filopodia, and focal adhesions are detected with the paxillin channel. (D) Composite mask (the area highlighted is magnified on the right) from all individual annotations is used in training. Scale bars: 10 μ m.

loss for 100 epochs, with a dataset split of 237 training images, 30 validation images and 30 test images, randomly selected across four experimental repeats (Fig. S3C, see Materials and Methods, ‘Training parameters’). Performance on validation images was used to select the model with the highest average Dice score of

0.716 (0.78 for actin and stress fibers, 0.55 for focal adhesions, 0.76 for lamellipodia and lamellar regions, 0.5 for filopodia, and 0.99 for background).

To evaluate the model performance, we analyzed test images unseen during training and compared the annotated mask and

predicted masks to get a comparable Dice score of 0.721 (0.78 for actin and stress fibers, 0.54 for focal adhesions, 0.78 for lamellipodia and lamellar regions, 0.51 for filopodia, 0.99 for background). A representative test image, along with the corresponding ground truth, predicted mask and overlay of the predicted mask on the corresponding test image (Fig. 2A–D), as well as magnified views of a selected area (Fig. 2E–H) are shown. To assess whether there is any bias of detecting one class of actin structure over another, we calculated the fraction of each substructure and plotted the distribution of them against the same analysis from the ground truth masks (Fig. 2I). All four substructures were found to be within the expected range between predicted fractions of cortical actin and stress fibers (0.61 ± 0.08 , mean $\pm$ s.d.), focal adhesions (0.08 ± 0.03), lamellipodia and lamellar regions (0.3 ± 0.08), and filopodia (0.01 ± 0.01) when compared with ground truth fractions of stress fibers (0.63 ± 0.08), focal adhesions (0.07 ± 0.03), lamellipodia and lamellar regions (0.29 ± 0.09), and filopodia (0.01 ± 0.01) with no statistically significant differences between the predicted and inference values

for stress fibers ($P=0.33$), focal adhesions ($P=0.3$), lamellipodia and lamellar regions ($P=0.59$), and filopodia ($P=0.42$). To determine whether these predicted masks can be used to quantify actin classes, OpenCV was used to count the number of detections for each actin class in the annotated masks and predicted masks. These counts were plotted across all the test images (Fig. 2J) to assess the correlation between predicted counts and ground truth counts. A positive correlation was observed in stress fibers (Pearson's correlation=0.72), filopodia (Pearson's correlation=0.52), lamellipodia and lamellar regions (Pearson's correlation=0.18), and focal adhesions (Pearson's correlation=0.75). In addition, to evaluate the accuracy of predicted masks, we computed f1 scores by comparing each predicted mask in the test dataset unseen by the model with its corresponding ground truth mask (Fig. 2K). A strong overlap was seen in actin and stress fibers (f1 score= 0.78 ± 0.04 ; mean $\pm$ s.d.), focal adhesions (f1 score= 0.51 ± 0.12), lamellipodia and lamellar regions (f1 score= 0.75 ± 0.11), and filopodia (f1 score= 0.52 ± 0.17). To assess experimental and biological variability, we performed an analysis of random images that

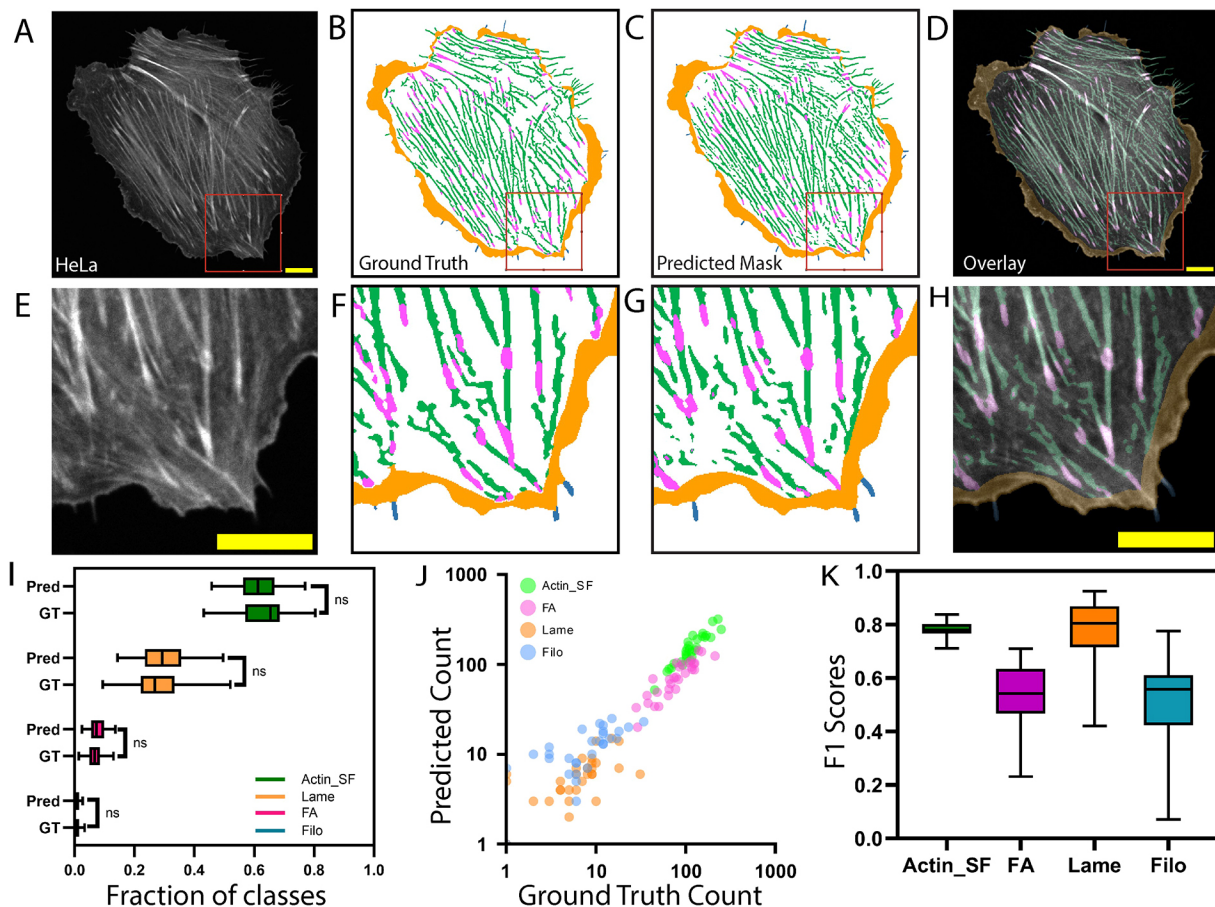


Fig. 2. FAST inference evaluation for actin structure prediction. (A) Representative image of HeLa cell from the phalloidin channel used as input for algorithm testing. (B) Corresponding annotated mask with actin and stress fibers (green), focal adhesions (red), lamellipodia and lamellar regions (yellow) and filopodia (blue). (C) Corresponding predicted mask from the FAST model. (D) The predicted mask is overlaid on input image to highlight the predicted structures. (E–H) Magnified views for highlighted regions of corresponding images. (I) For a random test set (of 30 images) chosen from four biological replicates, the distribution of fractional area of each subtype from predicted (Pred) masks is compared to that of the annotated masks (ground truth, GT) using a two-tailed unpaired *t*-test to show no significant differences for stress fibers (Actin_SF; $P=0.33$), focal adhesions (FA; $P=0.3$), lamellipodia and lamellar regions (Lame; $P=0.59$), and filopodia (Filo; $P=0.42$). (J) Scatter plot on a log-log scale of the count of individual substructures from prediction mask for test set image and corresponding ground truth mask showing correlation between predicted counts with annotated counts. (K) Plot showing f1 scores that indicate overlap between predicted masks and corresponding ground truth masks for stress fibers (f1 score= 0.78 ± 0.04), focal adhesions (f1 score= 0.51 ± 0.12), lamellipodia and lamellar regions (f1 score= 0.75 ± 0.11), and filopodia (f1 score= 0.52 ± 0.17). f1 scores are mean $\pm$ s.d., $n=30$. ns, not significant. For box plots, the box represents the 25–75th percentiles, and the median is indicated. The whiskers show the complete range. Scale bars: 10 μ m.

were reimaged under identical conditions (Fig. S4). No significant differences were observed between replicate imaging on the same cells after remounting the sample, indicating that our analysis tool was not influenced by the reproducibility of the imaging setup.

FAST analysis of different cell types

To evaluate the ability of FAST to accurately detect different classes of actin structure from images of different cell types, two additional commonly used adherent cell lines were selected for validation – the mouse NIH-3T3 fibroblast and the porcine LLC-PK1 epithelial cell line (see Materials and Methods, ‘Cell culture’). Imaging was independently performed three times, following the same procedure used for HeLa cells, and ten images were collected from each session, resulting in 30 multichannel images per cell line. A representative image from each of the cell lines, along with its corresponding annotated and predicted mask, are shown (Figs 3A–H and 4A–H). In total, 240 images across the four imaging channels were processed and annotated with the same procedure as described above.

All collected images were processed with the trained model, which had not seen these images or these cell types during training. To assess model performance, we calculated the fraction of each class of actin structure and plotted their distributions (Figs 3I and 4I). The predicted fractions for LLC-PK1 cells for actin and stress fibers (0.47 ± 0.09), focal adhesions (0.08 ± 0.02), and lamellipodia and lamellar regions (0.45 ± 0.1) closely matched the actin and stress fibers (0.49 ± 0.1), focal adhesions (0.08 ± 0.02), and lamellipodia and lamellar regions (0.44 ± 0.11) values for the ground truth fractions, with no statistically significant difference for actin and stress fibers ($P=0.38$), focal adhesions ($P=0.1$), and lamellipodia and lamellar regions ($P=0.69$) (Fig. 3I). Although there was a lack of filopodia in ground truth (0.002 ± 0.002) and predicted masks (0.003 ± 0.003), the LLC-PK1 cells showed a statistically significant difference ($P=0.004$) for filopodia. The predicted fractions for NIH-3T3 cells for stress fibers (0.55 ± 0.10), focal adhesions (0.07 ± 0.02), lamellipodia and lamellar regions (0.34 ± 0.09), and filopodia (0.02 ± 0.01) closely matched the cortical actin and stress fibers (0.6 ± 0.1), focal adhesions (0.07 ± 0.02), lamellipodia and lamellar regions (0.3 ± 0.09), and filopodia

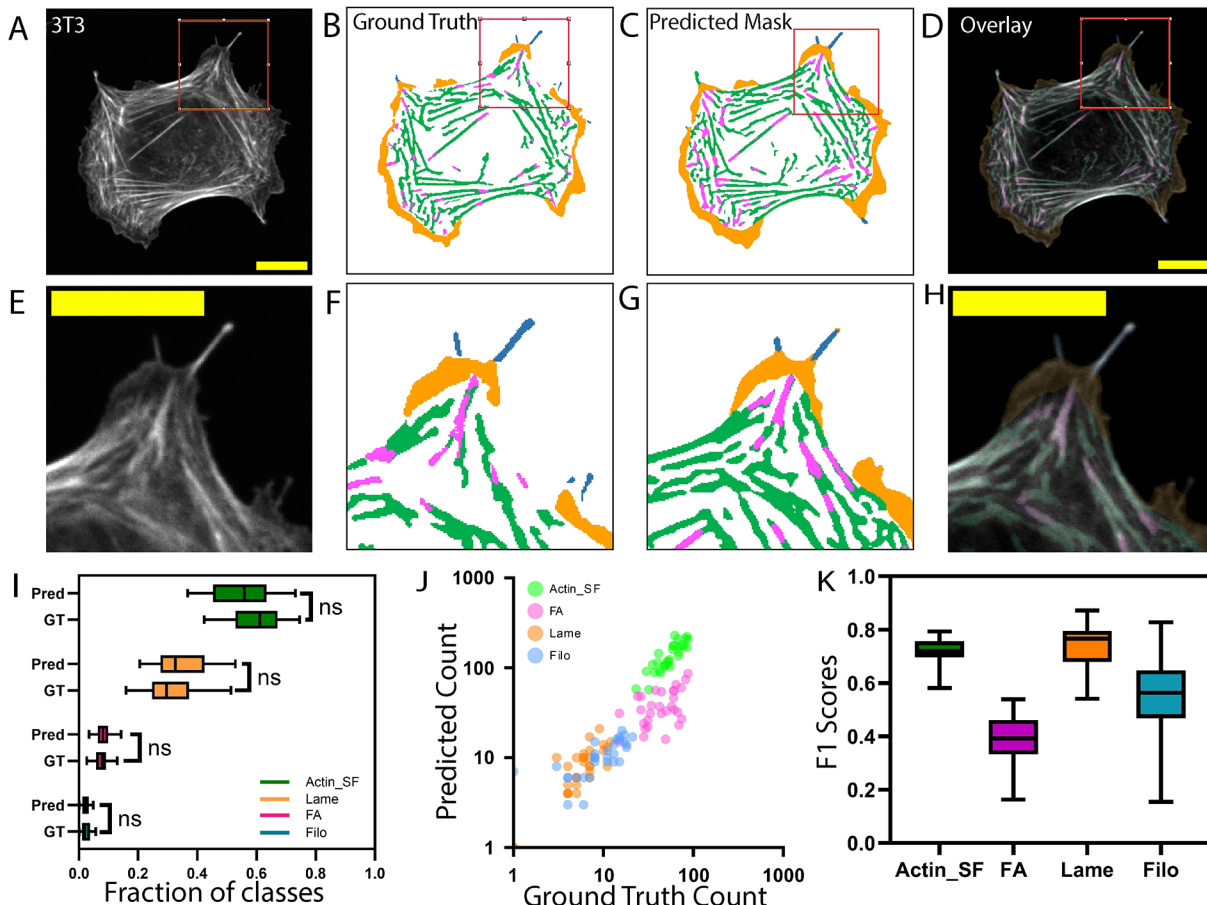


Fig. 3. FAST generalization evaluation on the alternative cell type LLC-PK1. (A) Representative image from LLC-PK1 cells with (B) the ground truth, and (C) predicted mask and (D) overlay of predicted mask on input image. (E–H) Magnified views for highlighted regions of corresponding images in A–D. (I) For a test set (of 30 images) chosen from three biological replicates with ten images per replicate, the distribution of fractional area of each subtype from predicted (Pred) masks is compared to that of the annotated masks (ground truth, GT) using a two-tailed unpaired *t*-test to show no significant differences for stress fibers (Actin_SF; $P=0.38$), focal adhesions (FA; $P=0.1$), and lamellipodia and lamellar regions (Lame; $P=0.69$) but a statistically significant difference was found for filopodia (Filo; $P=0.004$). (J) Scatter plot on a log-log scale of the count of individual substructures from prediction mask for test set image and corresponding ground truth mask showing correlation between predicted counts with annotated counts. (K) Plot showing f1 scores that indicate overlap between predicted masks and corresponding ground truth masks for stress fibers (f1 score= 0.74 ± 0.05), focal adhesions (f1 score= 0.51 ± 0.12), lamellipodia and lamellar regions (f1 score= 0.8 ± 0.06), and filopodia (f1 score= 0.37 ± 0.35). f1 scores are mean $\pm$ s.d., $n=30$. ns, not significant. For box plots, the box represents the 25–75th percentiles, and the median is indicated. The whiskers show the complete range. Scale bars: 10 μ m.

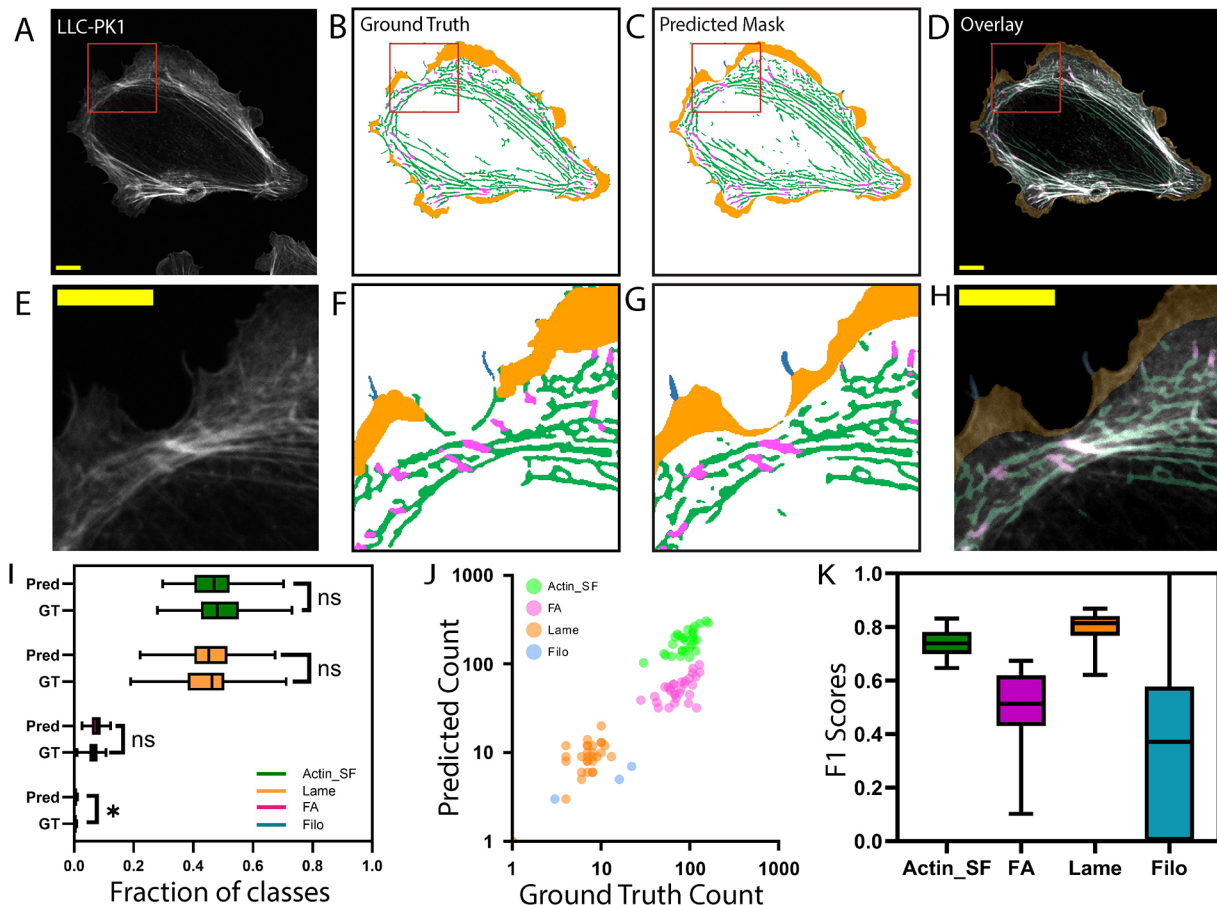


Fig. 4. FAST generalization evaluation on the alternative cell type NIH-3T3. (A) Representative image from NIH-3T3 with (B) the ground truth, and (C) predicted mask and (D) overlay of predicted mask on input image. (E–H) Magnified views for highlighted regions of corresponding images in A–D. (I) For a test set (of 30 images) chosen from three biological replicates with ten images per replicate, the distribution of fractional area of each subtype from predicted (Pred) masks is compared to that of the annotated masks (ground truth, GT) using a two-tailed unpaired *t*-test to show no significant differences for stress fibers (Actin_SF; $P=0.06$), focal adhesions ($P=0.17$), lamellipodia and lamellar regions (Lame; $P=0.12$), and filopodia (Filo; $P=0.99$). (J) Scatter plot on a log-log scale of the count of individual substructures from prediction mask for test set image and corresponding ground truth mask showing correlation between predicted counts with annotated counts. (K) Plot showing f1 scores that indicate overlap between predicted masks and corresponding ground truth masks for stress fibers (f1 score= 0.72 ± 0.05), focal adhesions (f1 score= 0.38 ± 0.1), lamellipodia and lamellar regions (f1 score= 0.74 ± 0.08), and filopodia (f1 score= 0.57 ± 0.14). f1 scores are mean $\pm$ s.d., $n=30$. * $P<0.05$; ns, not significant. For box plots, the box represents the 25–75th percentiles, and the median is indicated. The whiskers show the complete range. Scale bars: 10 μ m.

(0.02 ± 0.02) values for the ground truth fractions with no statistically significant differences between stress fibers ($P=0.06$), focal adhesions ($P=0.17$), lamellipodia and lamellar regions ($P=0.12$), and filopodia ($P=0.99$) (Fig. 4I). Interestingly, the fraction of lamellipodial spread was higher in LLC-PK1 cells (0.45 ± 0.1) in comparison to HeLa cells (0.3 ± 0.08 , $P=8.2\times10^{-8}$) and NIH-3T3 cells (0.34 ± 0.09 , $P=7.8\times10^{-7}$), which is a characteristic difference in morphology between these cell types.

The ground truth mask and predicted mask were further analyzed by counting the number of each class of actin structure per cell (Fig. 3J and 4J). This analysis on LLC-PK1 cells revealed no correlation in filopodia (Pearson's correlation= 0.02), due to absence of them, but strong correlation was seen in stress fibers (Pearson's correlation= 0.5), lamellipodia and lamellar regions (Pearson's correlation= 0.38), and focal adhesions (Pearson's correlation= 0.51). This was also seen in NIH-3T3 cells (Fig. 4J), with strong correlation in filopodia (Pearson's correlation= 0.64), stress fibers (Pearson's correlation= 0.72), lamellipodia and lamellar regions (Pearson's correlation= 0.51), and focal adhesions (Pearson's correlation= 0.32).

In addition, to evaluate the accuracy of predicted masks, we computed f1 scores by comparing each predicted mask in the test

dataset unseen by the model with its corresponding ground truth mask (Figs 3K and 4K). This analysis on LLC-PK1 cells revealed a significant overlap in stress fibers (f1 score= 0.74 ± 0.05), lamellipodia and lamellar regions (f1 score= 0.8 ± 0.06), focal adhesions (f1 score= 0.51 ± 0.12), and filopodia (f1 score= 0.37 ± 0.35). This was also seen in NIH-3T3 cells (Fig. 4K) with strong overlap in stress fibers (f1 score= 0.72 ± 0.05), lamellipodia and lamellar regions (f1 score= 0.74 ± 0.08), focal adhesions (f1 score= 0.38 ± 0.1), and filopodia (f1 score= 0.57 ± 0.14).

FAST analysis of live-cell imaging data

To test the ability of FAST to quantify dynamic changes in the organization of the actin cytoskeleton in live cells, we generated a stable cell line expressing LifeAct-GFP in NIH-3T3 fibroblasts (Fig. 5A) and analyzed with FAST (Fig. 5B–D). Upon treatment of these cells with Rho-associated kinase (ROCK) inhibitor Y-27632 (50 μ M), we observed notable differences in the abundance of different actin classes as seen in the same cell after 30 min of incubation (Fig. 5E) and analyzed with FAST (Fig. 5F–H). For population analysis, we collected 42 images across three biological replicates (10–20 images per replicate) of cells treated with ROCK

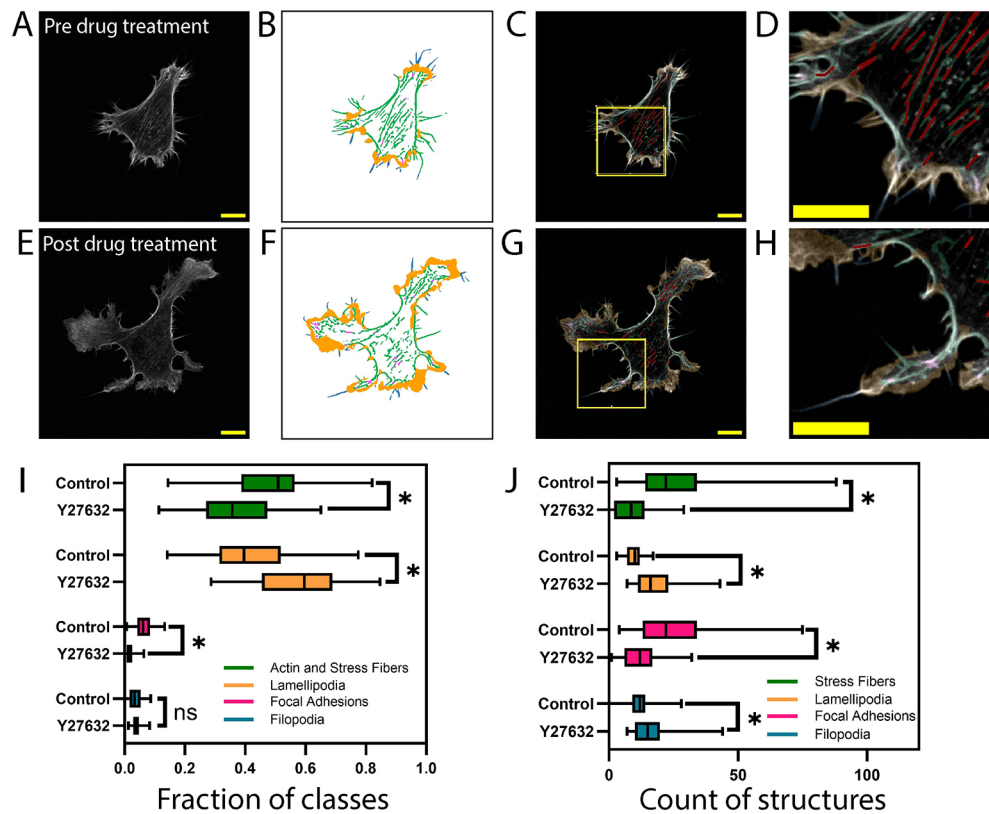


Fig. 5. FAST inference on live-cell imaging data. (A) NIH-3T3 fibroblasts expressing LifeAct-GFP were imaged prior to drug treatment and analyzed with FAST giving a predicted mask (B) that can be overlaid on the input image (C) along with its magnified regions (highlighted by yellow box) (D). Example showing the same cell was treated with Y27632 and imaged after 30 min (E) and analyzed with FAST giving a predicted mask (F) that can be overlaid on input image (G) along with its magnified regions (highlighted by yellow box) (H). (I) For a control set (of 42 images) with DMSO vehicle control chosen from three biological replicates, the distribution of fractional area of each subtype from predicted masks is compared to that of the predicted masks of Y27632-treated cells (of 40 images) from three biological replicates using a two-tailed unpaired *t*-test to show significant differences for actin and stress fibers ($P=4.3 \times 10^{-4}$), focal adhesions ($P=5.7 \times 10^{-12}$), and lamellipodia and lamellar regions ($P=3.6 \times 10^{-6}$). (J) The count of classes of actin structure that are detected from predicted masks of control cells showing statistically significant changes in the count of stress fibers ($P=1.2 \times 10^{-6}$), focal adhesions ($P=1 \times 10^{-5}$), lamellipodia and lamellar regions ($P=7.4 \times 10^{-7}$), and filopodia ($P=4.2 \times 10^{-4}$), when compared to Y27632 treated cells using two-tailed unpaired *t*-test. * $P < 0.05$; ns, not significant. For box plots, the box represents the 25–75th percentiles, and the median is indicated. The whiskers show the complete range. Scale bars: 10 μ m.

inhibitor Y-27632, and compared to 40 images across three biological replicates (10–20 images per replicate) of different cells with vehicle control DMSO (see Materials and Methods, ‘Cytoskeletal drug treatments’).

We calculated the fraction of each substructure and plotted the distribution for control DMSO cells and cells treated with Y-27632 (Fig. 5I). Box plots of each label count showed significant differences among the drug conditions, especially there was a significant increase ($P=3.6 \times 10^{-6}$) in the fraction of lamellipodia and lamellar regions from control (0.42 ± 0.14) to ROCK inhibitor (0.58 ± 0.14) along with significant reduction ($P=4.3 \times 10^{-4}$) in actin and stress fiber fraction from control (0.48 ± 0.14) to ROCK inhibitor (0.37 ± 0.14) and focal adhesion fraction ($P=5.7 \times 10^{-12}$) from control (0.06 ± 0.03) to ROCK inhibition (0.02 ± 0.01).

For quantitative analysis, we counted the number of filopodia, lamellipodial and lamellar networks, stress fibers, and focal adhesions under each condition with OpenCV contour counts (Fig. 5J). For stress fibers, we used a line segment detector (OpenCV) to count the fibers with actin label for the ground truth and predicted masks; the other substructures were counted with corresponding contours. Box plots of each label count showed significant differences among the drug conditions, especially there was a significant increase ($P=4.2 \times 10^{-3}$) in filopodia count from

control (11 ± 5) to ROCK inhibitor (16 ± 8), and lamellipodia count ($P=7.4 \times 10^{-7}$) from control (10 ± 3) to ROCK inhibitor (17 ± 8), along with a significant reduction ($P=1.2 \times 10^{-6}$) in stress fibers from control (25 ± 18) to ROCK inhibitor (8 ± 7), and focal adhesion count ($P=1 \times 10^{-5}$) from control (26 ± 17) to ROCK inhibition (12 ± 7).

DISCUSSION

In this work, we developed FAST to enable accurate and automated detection of four different classes of actin structure using phalloidin images and live-cell imaging with LifeAct-GFP, without needing additional antibodies. It can process thousands of images within an hour (>10 images per second) with reasonable sized GPUs and is well-suited for faster hardware acceleration with in-house and cloud computing if needed. The frozen model ensures consistent results for repeated runs of a given image free from human bias. Both the trained model and the labeled dataset are open source (see ‘Data and resource availability’), allowing researchers to use them as-is or fine-tune it further to meet specific needs. Although this tool was trained with data augmentation to account for variation within data, we found that the analysis accuracy largely depends on the quality of the image captured. Specifically, signal-to-noise ratio, contrast, extent of photobleaching and the focus of the cell play a vital role in prediction accuracy. Although low-density

plating of cells is common practice to avoid overlapping cells and limit the number of cells per image, images containing multiple cells can be cropped prior to processing to analyze the cells individually.

To assess whether our analysis could be applied to other cell types, we applied the trained FAST model to datasets from different adherent cell types beyond the ones used in training. FAST was still able to identify the four classes of actin structures with reasonable accuracy, though statistical confidence was lower than with the original cell types. Additionally, we used FAST to distinguish distinct morphological features present in different cell lines by quantifying the differences in the abundance of lamellipodia between LLC-PK1, HeLa and NIH-3T3 cells (Figs 2–4). We further highlighted an application of FAST by measuring changes in cytoskeletal organization during pharmacological treatments (Fig. 5) for quantification of drug-induced changes to cytoskeletal structures. Although FAST can be used across a range of cell types and treatments, its performance might benefit from finetuning on cell-type-specific data to help the model avoid misclassification of unseen classes of actin structure. In future work, further fine-tuning of FAST or the addition of other classes of actin structure, such as cortical actin, puncta, asters and stars, could improve the model performance and its widespread use for other cell types and treatments.

As there are multiple tools available for actin cytoskeleton analysis, the optimal choice of the tool ultimately depends on the volume of data and the types of antibodies or markers used. In cases where there are only a few images to be analyzed, manual approaches using tools like ImageJ might be sufficient. FAST is best suited to analyze large number of images for classes of actin structure from phalloidin-stained fixed cells and live-cell imaging of cells expressing a signal for F-actin. However, if the dataset includes markers specific to a single substructure, alternative tools might also be adequate. For example, for single class quantification of filopodia with anti-myosin X antibody, filoVision could be a more suitable option. We compared the predictions from FAST with two other tools including FiloQuant and Filament Sensor, and observed a positive correlation of predicted structures (Fig. S5). Note that direct comparison of all the structures assessed with the FAST prediction is challenging, as some tools, like filoVision, require specific antibody staining, whereas others might not distinguish filopodia from retraction fibers. A key value of FAST is that it generates segmentation masks that can form the basis for future improvements, including enhanced stress fiber detection and the development of more robust metrics.

To gain a comprehensive understanding of cellular processes, it is essential to analyze different classes of actin structure simultaneously – a feature lacking in most current tools. Additionally, these tools also need to be compatible with diverse cell types and live-cell imaging to be widely useful. Unfortunately, many of the existing tools are not well suited to quantify dynamic changes in live-cell imaging. This underscores the need for an actin quantification tool like FAST, which can simultaneously detect multiple structures from diverse cell lines and with live-cell imaging. Moreover, it is costly and time consuming to source different antibodies to detect each kind of actin substructure individually. FAST was developed to address the need for detection of classes of actin structure without multiple specific antibodies, making them incompatible to quantify dynamic changes. The antibody-assisted labeling approach implemented here has shown potential to train actin cytoskeleton quantification models.

MATERIALS AND METHODS

Cell culture

HeLa cells (ATCC CCL-2), NIH-3T3 wild-type fibroblasts (ATCC CRL-1658) and LLC-PK1 (ATCC CL-101) cells were cultured in Dulbecco's modified Eagle's medium (DMEM; Gibco, 11965118) supplemented with 10% fetal bovine serum (FBS; Thermo Fisher Scientific, 10270106) and 1% penicillin-streptomycin (PS; Cytiva HyClone, SV30010) solution. Cultures were maintained at 37°C in 5% CO₂ and passaged when confluent using 0.25% trypsin-EDTA solution. For experiments conducted on eight-well imaging chambers (CellVis), the chambers were coated with 10 µg/ml fibronectin (Thermo Fisher Scientific, CB40008) and incubated for 1 h at 37°C. Then the cells were plated at a lower density and incubated for 6 h in DMEM supplemented with FBS and PS.

For live-cell imaging experiments, an NIH-3T3 cell line expressing LifeAct-GFP was generated following protocols for developing stable cell lines described in Harris et al. (2020). Briefly, double stranded DNA encoding LifeAct-GFP was synthesized by Integrated DNA Technologies (g-block) and inserted into the lentiviral vector pHR using Gibson assembly. Lentivirus was generated in HEK293T cells by transfecting pHR LifeAct-GFP with the helper plasmids PMD2.G and P8.91, using transit 293 transfection reagent. After 24–48 h, lentiviral supernatant was collected, filtered using a 0.4 µm filter, and applied directly to 75% confluent NIH-3T3 cells in a six-well plate. Cells were confirmed to be stably expressing LifeAct-GFP by using fluorescence microscopy at 24–48 h after infection.

Antibodies and reagents

Primary antibodies used in this study include paxillin antibody with sheep host (Novus Biologicals, AF4259-SP), integrin β5 antibody (Novus Biologicals, AF3824-SP), myosin X antibody with mouse host (Novus Biologicals, NBP2-88926), anti-MYO10 antibody with rabbit host (Abcam, ab224120), lamellipodin antibody (H-5) with mouse host (Santa Cruz Biotechnology, sc-390050), lamellipodin (D8A2K) with rabbit host (Cell Signaling Technology, 91138T), anti-myosin IIA (Sigma-Aldrich, M8064-25UL), human myosin IIA (GeneTex, GTX33939), phospho-myosin light chain 2 (Ser19) antibody (Cell Signaling Technology, 3671T), Arp3 antibody (A-1) Alexa Fluor® 488 (Santa Cruz Biotechnology, sc-48344 AF488), anti-paxillin antibody [Y113] with rabbit host (Abcam, ab32084), and phalloidin-iFluor 647 reagent (Abcam, ab176759) for staining actin. Secondary antibodies used in this study include donkey anti-sheep IgG H&L (Alexa Fluor® 405) (Abcam, ab175676), goat anti-mouse IgG (H+L) (Alexa Fluor® 405) (Life Technologies Inc, A31553), goat anti-human IgG (H+L) (Alexa Fluor® 488) (Life Technologies Inc, A11013), donkey anti-mouse IgG H&L (Alexa Fluor® 488) (Abcam, ab150105) and goat anti-rabbit IgG H&L (Alexa Fluor® 568) (Abcam, ab175471). Both primary and secondary antibodies were diluted in blocking solution according to the manufacturer's protocol at a 1:200 dilution with 2 mg/ml bovine serum albumin (BSA) (Sigma-Aldrich, A9418) in phosphate-buffered saline (PBS) (Corning, 21-030-CV).

Cytoskeletal drug treatments

Pharmacological treatments involving cytoskeletal drugs were used as agonists to form actin subtypes. To obtain microscopy images of cells that had undergone cytoskeletal drug treatments, the cells were incubated in an eight-well chamber with DMEM supplemented with 10% FBS and 1% PS for 6 h. At 30 min prior to performing the immunostaining, the cytoskeletal drugs were added to DMEM at the following concentrations: 50 µM of Y-27632 dihydrochloride, ROCK inhibitor (Abcam, ab120129). For the control well, the DMEM was mixed with 1.5 µl of DMSO (Invitrogen, D12345) as a vehicle control. The cells were then prepared for imaging according to the protocol described below under 'Immunostaining' for a single channel.

Immunostaining

Cells were fixed in 4% paraformaldehyde in cytoskeleton-preserving buffer (80 mM PIPES pH 6.8, 5 mM EGTA, and 2 mM MgCl₂ in distilled H₂O) and 0.1 mg/ml sucrose for 15 min at 37°C. Cells were then washed three times with phosphate-buffered saline (PBS) and permeabilized with 0.1% Triton X-100 in PBS for 20 min at room temperature. Cells were again washed three times with PBS and subsequently blocked with

2 mg/ml BSA in PBS overnight at room temperature to prevent non-specific binding.

For single-channel imaging experiments that do not involve actin subtype antibody binding (like drug treatment), cells were then stained with phalloidin-iFluor 647 reagent (Abcam, ab176759) at a 1:200 dilution in 2 mg/ml BSA in PBS for 1 h at room temperature. Prior to imaging, the staining solution was removed from each well and rinsed twice with PBS and subsequently replaced with PBS containing 0.1% sodium azide (Sigma-Aldrich, 08591-1ML-F).

For multi-channel imaging experiments, after blocking the cells were then added with primary antibodies of paxillin with rabbit host (ab32084), myosin X with mouse host (CL8994) and human Myosin IIA (GeneTex, GTX33939) at a 1:200 dilution in 2 mg/ml BSA in PBS for 3 h at room temperature. This solution was removed from each well and rinsed three times with PBS and subsequently replaced with the secondary antibodies of goat anti-mouse IgG (H+L) (Alexa Fluor® 405) (Life Technologies Inc, A31553), goat anti-human IgG (H+L) (Alexa Fluor® 488) (Life Technologies Inc, A11013), goat anti-rabbit IgG H&L (Alexa Fluor® 568) (Abcam, ab175471), and phalloidin-iFluor 647 reagent (Abcam, ab176759) at a 1:200 dilution in 2 mg/ml BSA in PBS for 1 h at room temperature. Prior to imaging, the staining solution was removed from each well and rinsed twice with PBS and subsequently replaced with 0.1% sodium azide in PBS.

Imaging setup

Cells were imaged using Nikon Eclipse Ti2 inverted microscope utilizing a CrestOptics X-Light V3 spinning disk integrated with a Photometrics Kinetix camera with 60× Apo Plan water objective. Images were acquired using the NIS-Elements software v6.10.01 at a resolution of 1600 by 1600 pixels with pixel length of 0.108 µm. The exposure time for single-channel imaging of phalloidin was set as 200 ms with 5–10% laser power.

Training parameters

Full-sized grayscale images (of resolution 1600×1600) were used by input images with a batch size of 3. The dataset was randomly split into training, validation and test sets with 237-30-30 sizes. The UNet++ model was trained for 100 epochs on Nvidia A100 (40GB) with a learning rate of 0.001 using an Adam optimizer. Data augmentation was applied on each epoch by cropping (of resolution 800×800) at random locations, altering brightness (by factor between 0.5 to 1.5), and adding Gaussian noise (mean=0.0, s.d.=0.1) independently to roughly half of the images ($P=0.5$). The loss function used was Dice, which could be obtained from the Dice coefficient, which for a given sets A and B were defined as:

$$\text{Dice} = \frac{2 * |A \cap B|}{|A| + |B|}.$$

The best performing model on the validation set was selected and evaluated on test dataset as shown in the results.

Software tools used

Data was preprocessed using ImageJ distribution of Fiji v2.16.0 (Schindelin et al., 2012) for removing noise and reducing cytoplasmic signal. The images were semi annotated using Fiji/ImageJ and associated plugin FiloQuant (Jacquemet et al., 2019). Manual annotation of images was done using Supervisely (<https://supervisely.com/>, version 6.8.67, accessed in 2023), where each pixel was assigned a class value of zero to five where zero is (default) background, one corresponds to stress fibers and cortical actin, two corresponds to focal adhesions, three represents lamellipodia and lamellar regions, and four shows filopodial protrusions. The deep learning model was trained using PyTorch v2.6.0 on the annotated dataset. The training code along with a model inference pipeline was version controlled with GitHub and can be found at <https://github.com/Carleton-CTE-Lab/FAST>. Plots were prepared using PlotNeuralNet v1.0.0, GraphPad Prism v10.2.2, and Adobe Illustrator v29.3.1.

Statistical analysis

Data are presented as mean±s.d. Whiskers in the box plots indicate minimum and maximum values. Normality of the dataset was tested using

Shapiro–Wilk test and unpaired two-tailed t -tests with Welch's correction were performed to assess statistical significance between normally distributed samples. In cases where the distribution is not normal, the Mann–Whitney U -test was performed to assess statistical significance. In all cases, $P<0.05$ was considered significant. Models were evaluated using the Dice score (above) with 10-fold cross-validation, selecting the model with the best validation score. Additionally, f1 scores were reported to assess the model accuracy which for given sets A and B defined as:

$$f1_{score} = \frac{2 * P * R}{|P| + |R|}; \quad \text{where } P = \frac{|A \cap B|}{|B|} \text{ and } R = \frac{|A \cap B|}{|A|}.$$

Competing interests

The authors declare no competing or financial interests.

Author contributions

Conceptualization: A.R.H., V.A., J.C., A.G.; Data curation: V.A.; Formal analysis: V.A.; Funding acquisition: A.R.H., J.C., A.G.; Investigation: V.A.; Methodology: V.A.; Software: V.A., J.C., A.G.; Supervision: A.R.H.; Validation: V.A.; Writing – original draft: A.R.H., V.A.; Writing – review & editing: A.R.H., V.A., J.C., A.G.

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Data and resource availability

The filamentous actin segmentation tool (FAST) for quantifying cytoskeletal organization, including training and test datasets, is available at [doi:10.5281/zenodo.18135375](https://doi.org/10.5281/zenodo.18135375). A pretrained model for FAST (CC-BY 4.0 license) is available at [doi:10.5281/zenodo.18627454](https://doi.org/10.5281/zenodo.18627454). Source code for FAST (MIT license) is available on GitHub at <https://github.com/Carleton-CTE-Lab/FAST/>. All other relevant data and details of resources can be found within the article and its [supplementary information](#).

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